

THE ORBITAL ELECTRON IN THE TORSION MEDIUM

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doc_nucleus, doc_jobson_cell

ABSTRACT

The electron (E^+ spinor irrep of the $2I$ double group, vertex mode of the Jobson cell lattice) experiences four geometrically distinct forces when in orbit around a proton. This paper identifies each force, its physical mechanism, its torsionverse derivation status, and its contribution to orbital structure. The key result: the electron's roll direction (the chirality of its (1,2) Hopf winding) is not merely a labelling convention but determines the sign of its electromagnetic coupling to the proton. A positron (same E^+ irrep, opposite roll) is repelled from the proton for the same geometric reason that two protons repel. The four orbital forces together produce atomic shell quantization from I_h icosahedral geometry with zero free parameters.

Mass derivation is NOT in scope here -- see doc_leptons (LM1-LM2), doc_alpha (Born balance), doc_jobson_cell (J26). This paper covers orbital mechanics only.

Companion script: analysis/demos/electron_doc.py (16/16 PASS, ED1-ED16)

Companion doc: docs/series1/doc_orbit_pressure.txt (scale-invariant derivation of all orbital mechanics from torsion medium pressure gradients; PO1-PO8 PASS)

1. THE ELECTRON AS E^+ VERTEX MODE

1.1 Irrep assignment

The icosahedral double group $2I$ has four spinor irreps (half-integer spin):

E^+ (dim=2), E^- (dim=2), G_{32} (dim=4), I_{52} (dim=6)

The electron is assigned to E^+ on three grounds:

- (1) Spin-1/2: dim=2 spinor, consistent with measured g-factor [doc_alpha]
- (2) Vertex mode: E^+ winding scatters at the 12 icosahedral vertices.
 $\chi(E^+, C_5) = +\phi$ (constructive at 5-fold vertex nexuses).
 $\chi(E^+, C_3) = -1$ (color singlet: vertex where 5 colored faces meet).
- (3) CG selection: the V-A weak decay $T_{1g} \times E^- \rightarrow \{G_{32}, I_{52}\}$ excludes E^+ from pion decay products -- the electron is not produced in $\pi \rightarrow \mu + \nu$, only in the second-generation V-A chain. [doc_leptons Section 1.2]

E^- is the positron (E^+ winding, opposite roll -- see Section 4).

1.2 Cell interaction regime

$$N_{J_e} = \hbar c / (m_e \cdot L_J) = 38,870 \quad [\text{deep bulk}]$$

The electron is deep in the bulk regime ($N_J \gg 1$). It interacts with the Jobson cell lattice through vertex contacts (the icosahedral vertex nexuses), NOT through excluded-volume displacement. There is no Zone 1 or Zone 2 for the electron -- its displacement is topological (winding number 2 of the (1,2) Hopf torus), not geometric (no nuclear hard core on the attraction side).

This is why the electron-proton system has no hard-core repulsion: cells MESH at the electron-proton interface (complementary chirality), whereas two protons GRIND (same chirality, Zone 2 contact at $r = 2\lambda_p = 0.421$ fm).

[doc_nucleus Section 6; doc_orbit_pressure Section 1.2]

2. FORCES ON THE ORBITAL ELECTRON

The electron in orbit around a proton is subject to four geometrically distinct mechanisms. These are not separate fundamental forces -- all four emerge from the same torsion medium physics. The labels name where each effect is dominant.

2.1 PRESSURE (primary driving force)

MECHANISM — PROTON GRADIENT:

The proton's (1,2) Hopf winding in Zone 2 freezes Jobson cells into a specific chirality, expanding and thinning the medium in Zone 3. This creates a pressure gradient: Zone 3 cells are sparser (lower pressure) than the background. Any winding mode accumulates longer dwell times heading into the lower-pressure region and shorter dwell times heading out. The net drift is inward. This IS gravity and electromagnetism -- not two forces but one mechanism: dwell-time asymmetry in a pressure gradient. [doc_orbit_pressure Sections 1.1-1.2]

MECHANISM — ELECTRON SELF-WAKE:

The electron bumps into every Jobson cell it encounters -- it does not clear a path ahead. At each vertex contact ($\chi(E^+, C5) = \phi$, Born coupling α) the electron knocks the cell out of the way as it passes through. Displaced cells haven't snapped back yet, leaving:

- LOW pressure in the WAKE (tail) behind the electron: cells just passed through are displaced and sparse.
- HIGHER pressure ahead and to the sides: unperturbed cells the electron is about to hit, and cells pushed laterally.

The fractional displacement per contact IS α -- the fine structure constant quantifies how hard each collision displaces a cell (k_n/k_{eff} from the Born balance, doc_alpha J17/J24).

The orbital path curves inward because the proton's Zone 3 gradient makes the inward arc additionally sparse: cells are already thinned inward, so the electron accumulates longer dwell times going inward and is steered that way. The orbit is the electron perpetually following the inward-bending path of least resistance through unperturbed medium.

RESULT (Bohr radius):

$a_0 = \hbar^2 / (m_e \cdot c^2 \cdot \alpha_{em}) = 5.292e-11$ m [EXACT, zero free parameters]
[doc_orbit_pressure Section 1.3; P02 PASS]

DOMINANT over all other forces at atomic scales. Sets the orbital radius.

2.2 LIGHT LINGER (same pressure, wave-level readout)

MECHANISM:

The low-pressure wake trailing behind the electron is sparser than the surrounding medium. Waves propagating back into the wake travel

through this sparse region and linger there -- they take longer per cell hop than waves propagating forward (into the denser unperturbed medium ahead). This asymmetry means wave energy tends to remain in the wake behind the electron rather than radiate symmetrically in all directions.

Simultaneously, the proton's Zone 3 gradient makes the inward-bending wake sparser still: the proton's own thinning reinforces the wake along the inward arc. Light lingers even longer along the inward path than the outward path, producing the net inward phase drift that we call the Coulomb potential in QM.

NOTE: light linger is not separate from the pressure mechanism. It is what waves experience as they propagate through the pressure geometry -- the static description (pressure gradient) and the dynamic description (wave linger asymmetry) are two views of the same physical situation.

2.3 ROLL (chirality coupling, orbital angular momentum)

MECHANISM:

The Zone 2 Jobson cells are jammed (Maxwell-critical) and frozen in the proton's (1,2) Hopf chirality. They rotate with the proton. At each shared face boundary, the rotating Zone 2 cell drives the adjacent Zone 3 cell through direct contact at the triangular face edges -- the gluon antinodes at the shared edge midpoints act as the coupling interface, like cogs on the triangular face boundaries. This propagates outward cell-by-cell: each Zone 3 cell, once driven, drives its neighbor at the next shared face edge. The mechanism is geometric and mechanical. [cell_rotation_propagation.py RP1-RP4:
 $\phi * L_J = 2 * r_{mid}$ exactly (cells touch at antinodes), C3=+1 gluon coupling constructive, 72 deg forced by C5 geometry, propagates to all 12 vertices equally with zero attenuation]

The 12-vertex icosahedral geometry restricts which angular momentum states are stable: the electron's standing wave in the Coulomb well decomposes into I_h irreps. The orbital quantum number l maps to irrep dimension:
 $l=0$: A_g (1) s orbital
 $l=1$: T_{1g} (3) p orbitals
 $l=2$: H_g (5) d orbitals
 $l=3$: $T_{2g}+G_g$ f orbitals
 Degeneracy $2l+1$ is EXACT for $l=0..5$ from I_h character theory [doc_nucleus AS1]. Shell maxima $2n^2$ follow directly. Noble gas closings $Z=2,10,18,36,54,86$. [AS7]

At larger orbital radii (higher n), the Zone 3 co-rotation field weakens (it falls as $1/r^2$ with the pressure gradient). Higher shells have weaker roll coupling -- hence weaker spin-orbit effects for outer electrons.

2.4 WALL (inner boundary and shell counting)

INNER WALL (hard, established for all elements):

The nuclear Zone 1/2 boundary at $r \sim \lambda_p = r_p/4 = 0.21$ fm. Zone 2 cells are jammed (Maxwell-critical) and impenetrable to the electron's vertex-mode winding. This is a genuine physical boundary, not a potential -- the jammed cells physically cannot be entered. [doc_nucleus Section 6, ESSENTIALLY CLOSED]

SHELL COUNTING (from I_h irrep structure, not walls):

The number of electron states per shell is determined by the I_h irreducible representation structure of the Coulomb pressure well. This gives $2n^2$ exactly for all elements, Z-independent. [atomic_shells.py AS1-AS7, 7/7 PASS]

NOTE ON OUTER WALL: there is no established outer wall mechanism.

For a single proton ($Z=1$): the orbit radius is set entirely by the light-linger wave closure condition in the open Coulomb well -- no outer wall is needed or present.

For multi-electron atoms ($Z>1$): inner-shell electrons screen the nuclear charge, modifying the effective Coulomb gradient that the outer electron sees. This is a continuous potential modification, not a discrete wall. The outer electron's orbit is still set by the wave

closure condition in the (now-screened) effective potential.
The "outer wall" picture is not supported by existing scripts and is
not required to derive $2n^2$ shell maxima.

3. ROLL DIRECTION = CHARGE SIGN

3.1 The physical mechanism

The proton's (1,2) Hopf winding freezes Zone 2 cells into a specific
chirality. "Positive charge" is the label for this chirality direction:
the proton's inward winding. Two protons have the SAME chirality.

When two same-chirality sources approach, the cells between them are
dragged in OPPOSING directions. The cells grind. Grinding = high pressure
between them. Both sources flee the high-pressure zone. This is repulsion.

When opposite-chirality sources approach (proton + electron), the cells
between them are dragged in COMPLEMENTARY directions. They MESH: the cells
flow smoothly in one net direction (Bernoulli effect). Meshing = low pressure
between them. Both sources follow the low-pressure zone. This is attraction.

SAME chirality: GRINDING $\rightarrow$ high pressure $\rightarrow$ REPULSION
OPPOSITE chirality: MESHING $\rightarrow$ low pressure (Bernoulli) $\rightarrow$ ATTRACTION
[doc_orbit_pressure Section 1.2; doc_nucleus Section 6]

3.2 The electron is the proton's chirality complement

The electron has E+ winding with opposite roll to the proton's (1,2) Hopf
chirality. This opposite roll IS what we call "negative charge". It is not
a label assigned independently -- it follows directly from the cell-meshing
mechanism. The electron's roll direction is the unique E+ winding orientation
that produces attraction to the proton's Zone 2/3 cell chirality.

The electron does not CHOOSE to have negative charge. The 2I double group has
two E-type spinors: E+ and E-. The one that meshes with the proton is E+ (the
electron). E- is the positron. Both exist as equally valid 2I irreps. The
physical distinction is which one meshes with the proton's winding and which
one grinds. The proton selects the labelling by fixing the chirality reference.

3.3 Measured consequence: Coulomb energy at grinding contact

$r_{\text{grind}} = 2 \cdot \lambda_p = 0.421 \text{ fm}$ [Zone 2 boundaries touch; same chirality]
Coulomb energy at r_{grind} : $\alpha_{\text{em}} \cdot \hbar c / r_{\text{grind}} = 3.42 \text{ MeV}$
Observed nuclear hard core: 0.4-0.6 fm. Prediction: 0.421 fm. [MATCH]
[doc_nucleus Section 6, ESSENTIALLY CLOSED]

For the proton-electron (meshing) pair: no hard core exists. Attraction
is smooth all the way to Zone 1 (where the proton's impenetrable structure
finally prevents further approach). This asymmetry is why stable atoms exist.

4. POSITRON COMPARISON

4.1 Identity

The positron is E- (or equivalently: E+ with opposite roll). It is the 2I spinor conjugate of the electron. Same mass (CPT symmetry, automatic in the torsionverse from bilateral winding symmetry), same E+ vertex geometry, same C5 character (+phi), same Born balance. The sole physical difference: roll direction is reversed relative to the proton's Zone 2/3 chirality.

4.2 Positron near a proton

Roll: SAME as proton chirality (positron roll = proton roll direction).
Effect: GRINDING → high pressure between them → REPULSION.

The positron is repelled from the proton by the same cell-grinding mechanism that repels two protons. This is "like charges repel" -- now derived from geometric chirality, not assumed. The positron does not orbit the proton.

The Coulomb repulsion energy at r_{grind} for a positron:

$$\alpha_{\text{em}} * \hbar c / r_{\text{grind}} = 3.42 \text{ MeV} \quad [\text{same as proton-proton at contact}]$$

(The positron has no Zone 2, so the hard core at r_{grind} is softer than for proton-proton. The repulsion is present but less sharp at very short range.)

4.3 Positron near an electron (positronium)

Roll: OPPOSITE to electron → MESHING → ATTRACTION.

The positron-electron pair attracts by the same meshing mechanism as the proton-electron pair. Positronium (e+e- bound state) forms with Bohr radius:

$$a_{\text{ps}} = 2 * a_0 = 1.058 \times 10^{-10} \text{ m}$$

(twice a_0 because the reduced mass is $m_e/2$ vs the proton-electron case where $m_{\text{reduced}} \approx m_e$). This follows from the same pressure-gradient orbit formula with the reduced mass substitution.

Neither the electron nor the positron has a Zone 1 or Zone 2. There is no inner nuclear wall. When the e+e- pair falls to minimum separation ($r \rightarrow 0$ in the classical picture), their forward and backward E+ windings cancel directly: forward winding + backward winding = zero net winding → A_g (totally symmetric isotropic emission) = two back-to-back photons. Clean, complete, no Zone 1 wall to negotiate.

Positronium lifetime:

$\tau = 2\hbar / (m_e c^2 * \alpha^5) = 124.5 \text{ ps}$ (para-Ps, singlet)
PDG: 125.1 ps. Error: 0.5% [ED14 PASS, leading-order formula]
Physical: rate at which opposite-roll E+ windings overlap and cancel.
Full geometric derivation from E+ winding cancellation not yet done.

5. THE $Z > 137$ COULOMB LIMIT

5.1 Mechanism

The 1s Bohr radius is $a_{1s} = a_0 / Z$. As Z increases:

Z = 1: $a_0 = 52,918 \text{ fm} = 5.292 \text{ Angstrom}$ (hydrogen)
Z = 100: $a_0/100 = 529 \text{ fm}$ (still $\gg r_p = 0.841 \text{ fm}$; outside nucleus)
Z = 137: $a_0/137 = 386 \text{ fm} = \lambda_e$ (electron Compton wavelength)

At $Z = 1/\alpha = 137.036$, the 1s orbital radius exactly equals the electron's own Compton wavelength ($\lambda_e = \hbar c/m_e = 386 \text{ fm}$). This is an EXACT algebraic identity: $a_0 \alpha = \hbar c/m_e = \lambda_e$. [ED7-ED8, PASS]

Below λ_e the electron's (1,2) winding circuit cannot close -- the circuit is shorter than the electron's own spatial extent. The orbit collapses. The result is electron capture: $p + e^- \rightarrow n$ (the proton takes the E^+ winding into Zone 1, converting it to a d-quark mode).

NOTE: the Jobson cell scale $L_J = 9.93e-3 \text{ fm}$ is NOT the relevant UV cutoff here. L_J is 40,000x smaller than λ_e . The atomic system's UV limit is set by the electron's Compton wavelength, not the cell size. $Z_{\text{crit}} = 1/\alpha = 137.036$ is the cell-geometry limit because α is the ratio of the electron's Born coupling to the orbital wave condition:

$$a_0 = \lambda_e / \alpha \Rightarrow Z_{\text{crit}} = a_0 / \lambda_e = 1/\alpha.$$

[doc_nucleus Section 6.3; electron_doc.py ED7-ED9, ESSENTIALLY CLOSED]

5.2 Nucleus of the electron paper

This limit exposes the deepest connection: the fine structure constant α is not only the electron's Born coupling strength -- it is the ratio $(L_J / a_0) \times (\text{some geometry factor})$, i.e., the ratio of cell scale to orbital scale. The atomic system lives at scales $L_J \ll r \ll a_0$, i.e., in the window $1 \ll N_{J_{\text{orbital}}} \ll N_{J_e} = 38870$. At $Z = 137$, the orbital collapses to the cell scale, erasing this window. α 's numerical value SET the window in which stable atoms can exist.

6. SUMMARY: THE FOUR FORCES AND THE ROLL QUESTION

FORCE COMPARISON (hydrogen $Z=1$, ground state $E_{\text{H}} = -13.6 \text{ eV}$ as reference):

Force	Torsionverse source	Role	%
Pressure	Zone 3 pressure gradient (Coulomb well from proton)	Creates the potential well -- without it, no orbit is possible	~100% binding energy
Light linger	Wave asymmetry in the same pressure gradient (wave closure condition)	Selects WHICH radius is stable: the Bohr radius IS a linger-set resonance, not a classical force balance. Lamb shift (QED) = 0.00003% correction beyond this first-order closure.	~100% orbital position
Roll	Zone 2→Zone 3 cog coupling	Sets orbital SHAPE	~0.001%

	via gluon-antinode contact	(l, m_l): fine structure splitting per shell	energy
Wall	Zone 1/2 hard boundary + I_h irrep counting (AS1-AS7)	Inner: prevents $r \rightarrow 0$ Counting: $2n^2$ per shell	0% energy counting

NOTE: Pressure and light linger are inseparable. Pressure creates the gradient; linger is how a wave particle reads that gradient to select a stable radius. The Bohr radius = the radius where the electron's winding circuit closes exactly once in one orbital period (the wave resonance condition in the pressure-modified medium). Classical force balance alone does not determine this -- the wave nature does. Both are 100% contributors to different aspects of the same orbital.

ROLL = CHARGE SIGN:

Electron (E+, meshing with proton) = negative charge: DERIVED [Section 3]
Positron (E-, grinding with proton) = positive charge: DERIVED [Section 4]

POSITRON vs ELECTRON DIFFERENCES:

Mass: Identical (CPT, from bilateral winding symmetry) [CLOSED]
Charge: Opposite (roll direction relative to proton) [CLOSED, Section 3-4]
Orbit: Positron does NOT orbit proton (grinds); electron does (meshes)
Annihil: Positron + electron $\rightarrow$ A_g (photons, clean, no Zone 1 wall) [CLOSED]
Antiproton + proton $\rightarrow$ Zone 1 gluons released as pions (messy) [Section 4]

THE ROLL ANSWER:

The electron must roll the way it does relative to the proton because its E+ winding (forward roll) is the specific 2I spinor whose chirality complements the proton's (1,2) Hopf Zone 2 winding. If the electron had the opposite roll (= positron), it would not form a stable orbit around the proton at all. The universe selects which E-type spinor is "electron" and which is "positron" by the proton's chirality fixing the reference. CP violation (slight asymmetry in matter vs antimatter abundance) may trace to a slight imbalance in the initial population of E+ vs E- roll states in the cosmological medium. [SPECULATIVE, not yet derived]

7. GENUINELY OPEN ITEMS

INNER-SHELL SCREENING [MEDIUM]:

What is needed: derive the effective pressure cancellation fraction that one E+ winding (at orbital radius r_1) exerts on a second E+ winding at radius r_2 . The Slater coefficients (0.35 same-shell, 0.85 for n-1 shell, 1.00 for lower shells) are empirical fits to the cancellation ratio at each orbital separation. To derive them from torsionverse: model the pressure gradient of one electron winding as seen by another at a different orbital radius, then find the Z_{eff} that minimizes total torsion-medium energy. This is a two-E+-winding pressure calculation, not yet attempted.

CP VIOLATION [REFRAMED – LOCAL CHIRALITY, NOT UNIVERSAL]:

The torsionverse medium is built from light: the (1,2) Hopf winding is a light-mode lock. When Zone 2 first froze in our local region, it locked into one of two equally valid chiralities. Which chirality it chose was decided by whatever local fluctuation triggered the first freeze – not by a fundamental universal asymmetry.

Consequence: the matter-antimatter imbalance is our LOCAL condensation bias, not a property of the universe as a whole. Causally-disconnected regions that condensed with the opposite chirality would be entirely antimatter-dominated. We cannot easily detect them: light from antimatter stars is indistinguishable from matter-star light. The boundary between regions would show a pion annihilation zone – a gamma-ray excess along certain directions.

The observed $\eta \approx 6 \times 10^{-10}$ (baryon-to-photon ratio) may therefore not be a number to derive from first principles – it reflects the accident of our local freeze. The torsionverse "predicts" only that η is nonzero and region-dependent; its exact value is initial-condition noise.

What remains: can we derive a DISTRIBUTION of expected η across regions from the statistics of local freeze events? Connect to doc_redshift_time.txt (cell-generation rate, cosmological cell condensation mechanism). [SPECULATIVE]

CHIRALITY CASCADE (partially understood):

The local proton chirality is not chosen at the atomic scale -- it inherits from a cascade of medium currents: cosmic web filament → galactic center (depleted zone, pressure-gradient driver of galactic rotation) → solar nebula spin → Zone 2 freeze. The galactic center acts as a chirality AMPLIFIER: once a rotation sense is established at galaxy scale it propagates coherently downward via the near-incompressible medium ($K/G = 30.25$) to the Zone 2 freeze of individual protons. The chain terminates at the cosmic web scale, where the original filament angular momentum was set by quantum fluctuations in the initial medium condensation -- random in sign, region by region. No universal primary driver exists; the chirality of our matter is inherited from our cosmic web filament's accident of formation.

CLOSED THIS SESSION (ED15-ED16):

- Spin-orbit coupling: $\Delta E(2p) = \alpha^4 m_e c^2 / 32 = 45.3 \mu\text{eV}$, measured $45.35 \mu\text{eV}$, 0.15% [ED15 PASS]
- g-factor / Zone 3 effect on electron: the proton Zone 3 co-rotation couples to the electron spin via the effective magnetic moment ($g_p = 5.5857$), producing the hydrogen hyperfine splitting (21 cm line). Computed: $(4/3) \alpha^4 m_e (m_e/m_p) g_p = 1421.0 \text{ MHz}$, measured 1420.4 MHz , 0.04% [ED16 PASS]. Zone 3 co-rotation does NOT modify the free-space g-factor (2.00232); it creates the spin-energy splitting, which is already captured.